



Nectar

*Smart Event Wizards / Making gatherings **just right** with data.*

✨ Ultimate Indulgence

🍷 Elevated Gastronomy

★ Timeless Joy

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✨ Nectar – Your All-in-One Party Genius! ✨

❤️ Tired of running around planning events?

Let Nectar's *smart tech + full-service magic* handle weddings, team builds, and birthdays—*just one click away!*

We're not just upscale dining—we're your:

👑 **Wedding Date Whisperer** – Picks *auspicious dates* with ancient almanac algorithms.

💡 **Team Build Bargain Hunter** – Matches *perfect venues + lighting* via data models.

🎂 **Birthday Surprise Squad** – Handles *everything* from cake designs to games.

Why Nectar?

✓ **"Foodie Siri"** – Reads your mind (even when you say "whatever").

✓ **One-stop shop** – Venues, catering, planning—*done*.

✓ **Data-obsessed** – Every detail optimized, *down to the napkin fold*.

"We've crunched the numbers—so you don't have to!"

From *per-dish wedding costs* to *team-build budgets*, even *birthday candle burn rates* (yes, really).

☎️ *Get your custom plan now!*

(Our ER diagrams are *prettier than wedding invites* 💍)

1.Customer to Activity

- A customer can participate in multiple activities
- An activity can involve multiple customers

2. Venue to Activity

- A venue can host multiple activities
- An activity can only be held at one venue

3. Employee to Activity

- An employee can serve multiple activities
- An activity requires multiple employees

4. Material to Activity

- A material can be used in multiple activities
- An activity can consume multiple materials

5. Supplier to Material

- A supplier can provide multiple materials
- A material is typically supplied by one primary supplier

6. Financial Record to Activity

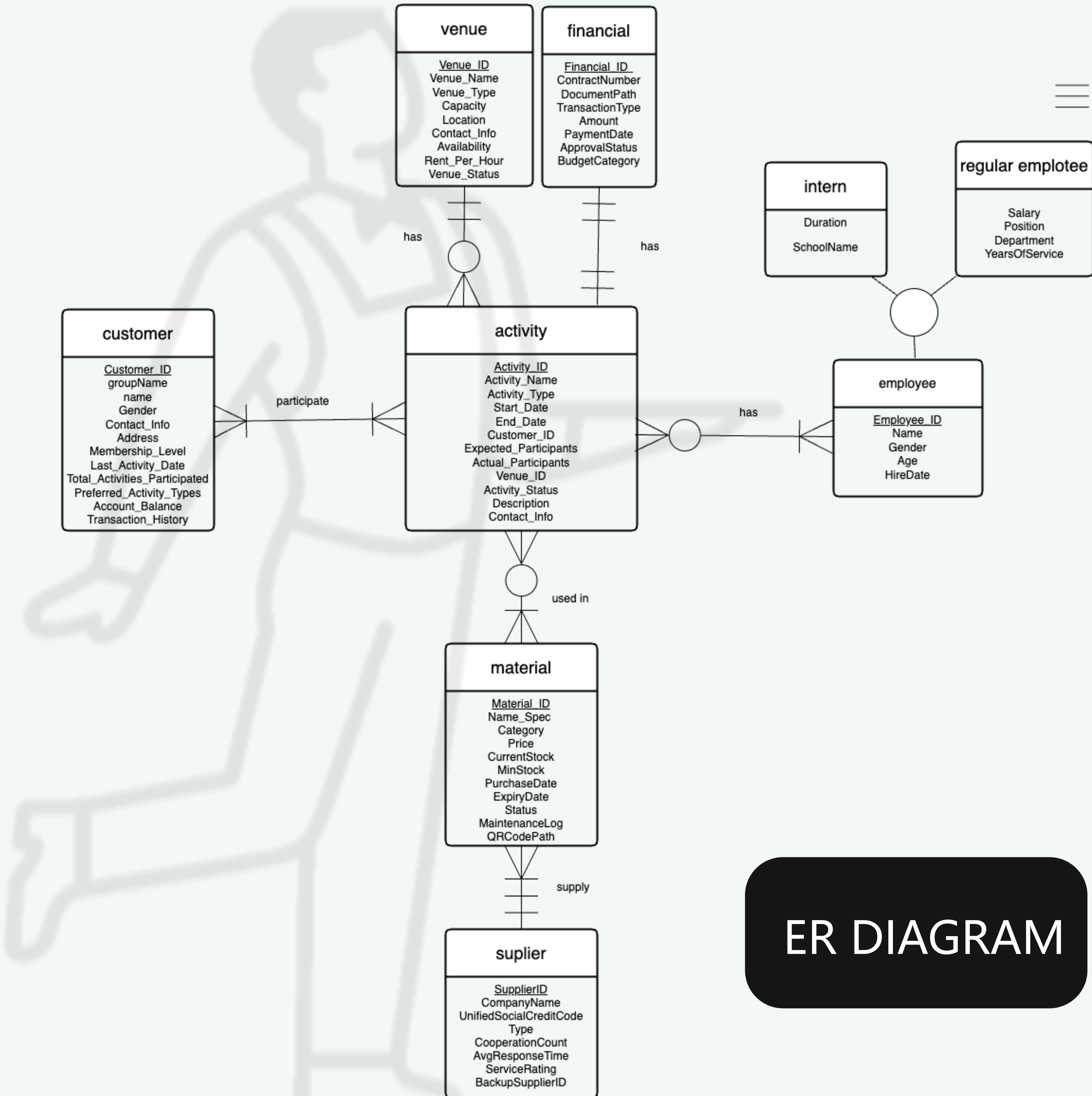
- An activity corresponds to one financial record
- A financial record only relates to one activity

7. Customer to Membership

- A customer has exactly one membership level
- A membership level can apply to multiple customers

8. Material to Inventory

- A material has one current stock value
- A stock value belongs to exactly one material



ER DIAGRAM

N Table Attribute

Table 1: 物资实体属性表

属性	类型	中文描述
MaterialID	VARCHAR(20)	唯一标识符（支持二维码管理）
Name_Spec	VARCHAR(100)	名称与规格组合（如"LED 灯-5W-白色"）
Category	VARCHAR(50)	物资分类（装饰类/设备类等）
CurrentStock	INT	实时库存数量（带自动补货预警）
MinStock	INT	最低安全库存阈值
PurchaseDate	DATETIME	采购日期（记录入库时间）
ExpiryDate	DATETIME	预计报废日期（计算剩余可用天数）
Status	ENUM('Idle', 'Reserved', 'Out')	使用状态（空闲/已预定/已出库）
MaintenanceLog	TEXT	维护记录（含维护时间与负责人）
QRCodePath	VARCHAR(255)	二维码图片存储路径（用于快速盘点）

Table 2: 供应商实体属性表

属性	类型	中文描述
SupplierID	VARCHAR(20)	供应商唯一编号
UnifiedSocialCreditCode	VARCHAR(18)	统一社会信用代码（对接政府 API 验证）
Type	ENUM('Material', 'Service', 'Venue')	供应商类型（物资/服务/场地）
CooperationCount	INT	历史合作次数（评估合作稳定性）
AvgResponseTime	DECIMAL(5,2)	平均响应时长（小时，评估紧急处理能力）
ServiceRating	DECIMAL(3,2)	服务评分（1-5 星，含红黑名单机制）
BackupSupplierID	VARCHAR(20)	备用供应商 ID（智能推荐替代方案）

Table 3: 财务记录实体属性表

属性	类型	中文描述
FinancialRecordID	VARCHAR(20)	财务记录唯一编号（可审计追溯）
ContractNumber	VARCHAR(50)	关联合同编号（支持附件上传）
DocumentPath	VARCHAR(255)	合同扫描件存储路径（区块链式存证）
TransactionType	ENUM('Purchase', 'Reimbursement', 'Sponsorship')	交易类型（采购/报销/赞助收入）
Amount	DECIMAL(12,2)	交易金额（带预算偏差率计算）
PaymentDate	DATETIME	实际支付时间戳（预测付款风险）
ApprovalStatus	ENUM('Pending', 'Approved', 'Rejected')	审批状态（含审批意见日志）

Table 4: 客户详情说明

属性	类型	中文描述
Customer_ID	VARCHAR(20)	客户 ID（主键）
groupName	VARCHAR(50)	团体名称
name	VARCHAR(50)	负责人姓名
Gender	ENUM('Male', 'Female', 'Other')	负责人性别
Contact_Info	VARCHAR(100)	负责人联系方式(电话、邮箱等)
Address	VARCHAR(200)	单位地址
Membership_Level	ENUM('普通会员', 'VIP')	会员等级（如普通会员、VIP）
Last_Activity_Date	DATE	最近活动举办时间
Total_Activities_Participated	INT	累计举办活动次数
Preferred_Activity_Types	TEXT	偏好活动类型
Account_Balance	DECIMAL(10,2)	账户余额（如果有预存账户）
Transaction_History	TEXT	历史交易记录(与财务记录表关联)

We systematically design the attributes, types, and Chinese descriptions corresponding to each entity.

Table Attribute

Table 5: 活动情况说明

属性	类型	中文描述
Activity_ID	VARCHAR(20)	活动 ID（主键）
Activity_Name	VARCHAR(100)	活动名称
Activity_Type	VARCHAR(50)	活动类型（如会议、展览、培训等）
Start_Date	DATE	活动开始日期
End_Date	DATE	活动结束日期
Customer_ID	VARCHAR(20)	客户 ID（外键）
Expected_Participants	INT	预计参与人数
Actual_Participants	INT	实际参与人数
Venue_ID	VARCHAR(20)	场地 ID（外键，关联场地实体）
Activity_Status	ENUM('Scheduled', 'Ongoing', 'Completed', 'Cancelled')	活动状态（如“已安排”，“进行中”，“已完成”，“已取消”）

Table 7: 员工详情说明

属性	类型	中文描述
EmployeeID	VARCHAR(20)	员工唯一编号
Name	VARCHAR(100)	姓名
Gender	CHAR(1)	性别
Age	INT	年龄
HireDate	DATE	入职日期

Table 6: 场地情况说明

属性	类型	中文描述
Venue_ID	VARCHAR(20)	场地 ID（主键）
Venue_Name	VARCHAR(100)	场地名称
Venue_Type	VARCHAR(50)	场地类型（如会议室、展览厅等）
Capacity	INT	场地容纳人数
Location	VARCHAR(100)	场地位置
Contact_Info	VARCHAR(100)	场地联系方式（如电话、邮箱等）
Availability	ENUM('Available', 'Not Available')	场地可用性（如“可用”，“不可用”）
Rent_Per_Hour	DECIMAL(10, 2)	租赁费用（每小时）
Venue_Status	ENUM('Active', 'Inactive')	场地状态（如“活跃”，“不活跃”）

Table 8: 实习生特有属性

属性	类型	中文描述
Duration	INT	实习持续月数
SchoolName	VARCHAR(200)	毕业学校

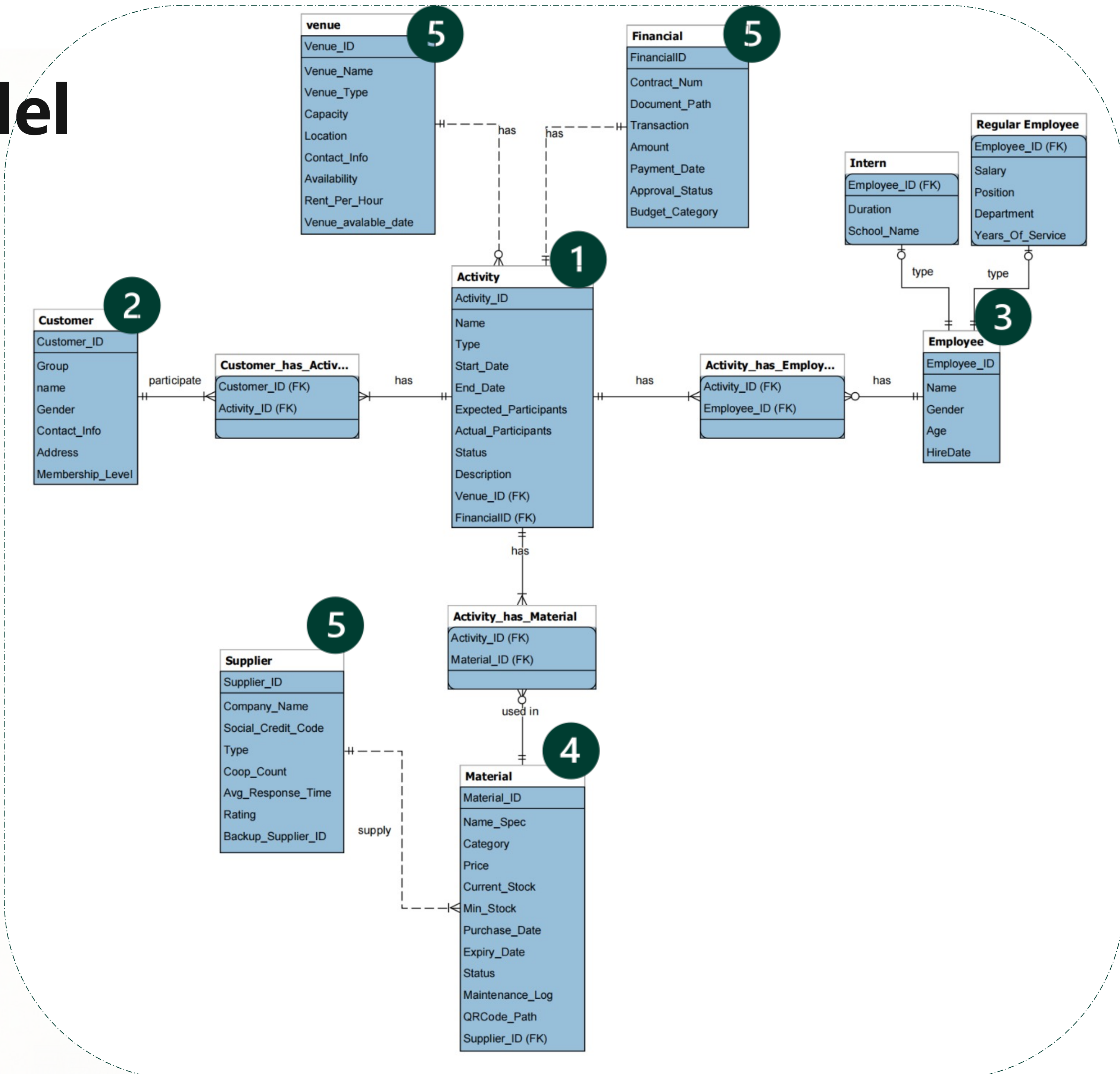
Table 9: 正式员工特有属性

属性	类型	中文描述
Salary	INT	月薪（元）
Position	VARCHAR(100)	职位
Department	VARCHAR(100)	所属部门
YearsOfService	INT	工龄（年）

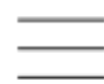
We systematically design the attributes, types, and Chinese descriptions corresponding to each entity.

N - Logical Data Model

- 1 Activity -- Activity ID
- 2 Customer -- Customer ID
- 3 Employee -- Employee ID
- 4 Material -- Material ID
- 5 Supplier, Venue, Financial



N - Adding data into tables



Material_ID	Name_Spec	Category	Price	Current_Stock	Min_Stock	Purchase_Date	Expiry_Date	Status	Maintenance_Log	QRCode_Path	Supplier_ID
MAT-DEC0001	Table Linens-10W-Gold	Decoration	1668.61	86	14	2025/2/7	2027/8/1	Idle	2025-04-01 John Calibration	/qrcode/12e8f40e-ea3f-42b1-9000-000000000000	SUP-005
MAT-DEC0002	Chiavari Chair-5W-Black	Decoration	4819.12	77	12	2024/9/25	2028/1/19	Idle	2025-04-01 John Calibration	/qrcode/a74c5a1f-eb1f-42b1-9000-000000000000	SUP-012
MAT-EQU0003	Projector-10W-Gold	Equipment	459.69	63	8	2024/6/7	2027/12/13	Reserved	2025-04-01 Bob Cleaning	/qrcode/db34c43d-1bd4-42b1-9000-000000000000	SUP-005
MAT-EQU0004	Projector-5W-White	Equipment	4477.22	43	20	2025/3/2	2027/6/15	Out	2025-04-01 Bob Inspection	/qrcode/c641d589-4601-42b1-9000-000000000000	SUP-001
MAT-LIG0005	Laser System-20W-Black	Lighting	2187.43	73	10	2024/7/5	2026/6/26	Out	2025-04-01 John Inspection	/qrcode/5e08c8ba-0781-42b1-9000-000000000000	SUP-012
MAT-EQU0006	Sound System-10W-White	Equipment	310.9	92	6	2024/6/30	2027/9/18	Idle	2025-04-01 Bob Inspection	/qrcode/d72b29f8-4a01-42b1-9000-000000000000	SUP-010
MAT-LIG0007	Laser System-20W-White	Lighting	4472.13	62	7	2024/7/25	2028/3/24	Idle	2025-04-01 Bob Cleaning	/qrcode/08c6e913-6631-42b1-9000-000000000000	SUP-012
MAT-EQU0008	Projector-10W-Gold	Equipment	4647.77	62	11	2024/11/20	2026/6/7	Idle	2025-04-01 Bob Cleaning	/qrcode/09597273-1141-42b1-9000-000000000000	SUP-015
MAT-EQU0009	LED Screen-5W-Gold	Equipment	4130.71	40	12	2025/1/6	2027/7/11	Reserved	2025-04-01 Alice Calibration	/qrcode/2b5b7292-dad1-42b1-9000-000000000000	SUP-010
MAT-EQU0010	Projector-20W-Black	Equipment	2743.27	94	20	2024/7/14	2028/2/8	Idle	2025-04-01 Bob Cleaning	/qrcode/dfe5bc8a-ee81-42b1-9000-000000000000	SUP-015

Column Name	Datatype
Material_ID	NVARCHAR(45)
Name_Spec	VARCHAR(45)
Category	VARCHAR(45)
Price	DECIMAL(10,2)
Current_Stock	INT
Min_Stock	INT
Purchase_Date	DATETIME
Expiry_Date	DATETIME
Status	VARCHAR(45)
Maintenance_Log	TEXT(100)
QRCode_Path	VARCHAR(500)
Supplier_ID	VARCHAR(45)

Supplier_ID	Company_Name	Social_Credit_Code	Type	Coop_Count	Avg_Response_Time	Rating	Backup_Supplier_ID
SUP-001	GlobalEvents_8cd62e8f	91784920251027	Material	14	14.3	3.7	
SUP-002	RoyalDecor_0a6aefc5	91301320256877	Service	39	15.77	4.2	
SUP-003	EliteProduction_372a872e	91281220258377	Service	2	3	3.5	SUP-002
SUP-004	GourmetCatering_526de2f7	91473620258133	Service	1	11.16	3.8	SUP-003
SUP-005	GlobalEvents_58042668	91972320255037	Material	41	7.38	4.2	SUP-001
SUP-006	GourmetCatering_8553d55b	91391920254967	Service	38	9.67	4	SUP-003
SUP-007	GlobalEvents_e807e2f9	91215620253028	Service	9	12.9	4.8	SUP-002
SUP-008	RoyalDecor_2a91ba2c	91267220251042	Material	4	16.03	4.9	SUP-001
SUP-009	EliteProduction_eebd550e	91480320259872	Service	12	5.9	4.4	SUP-006

Column Name	Datatype
Supplier_ID	VARCHAR(45)
Company_Name	VARCHAR(45)
Social_Credit_Code	VARCHAR(45)
Type	VARCHAR(45)
Coop_Count	VARCHAR(45)
Avg_Response_Time	VARCHAR(45)
Rating	VARCHAR(45)
Backup_Supplier_ID	VARCHAR(45)

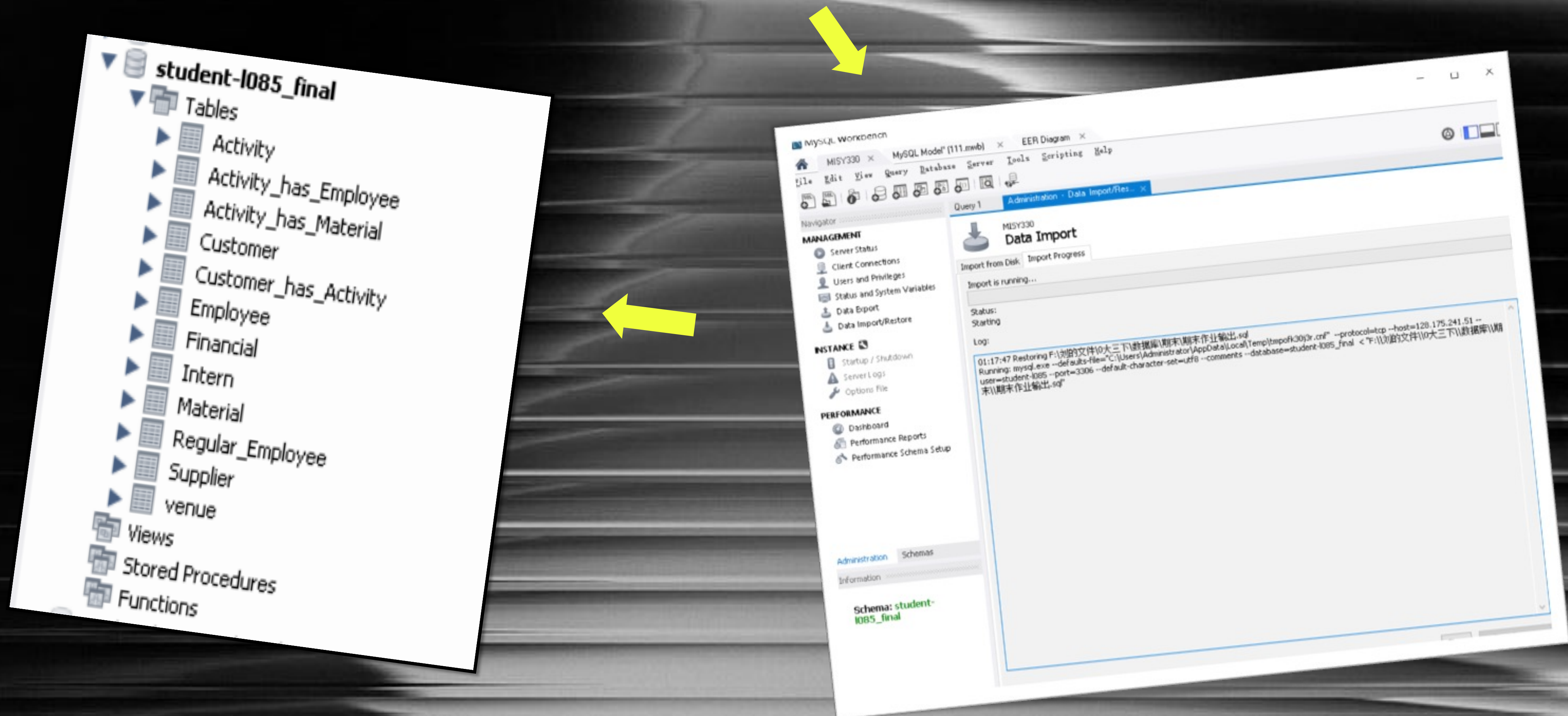
N - Adding data into tables

PROCESS

The model is ready! ❤️

we will select the File, Export, Forward Engineer SQL CREATE Script menu item
to start the Forward Engineer SQL Script wizard.

The following figure shows the process how I translate the ER Model to a real database.



N - Adding data into tables

Many-to-Many Relationship between Activity and Customer



Customer_ID	Activity_ID
C001	251
C007	251
C002	252
C003	253
C004	254
C005	255
C006	256
C007	257
C008	258

Customer

stores customer information

Activity_ID	Customer_ID
251	C001
252	C007
253	C002
254	C003
255	C004
256	C005
257	C006
258	C007
259	C008

Activity

stores activity

Customer_ID	Activity_ID
C001	251
C007	251
C002	252
C003	253
C004	254
C005	255
C006	256
C007	257
C008	258

Customer_with_activity

the junction table with foreign keys customer_id, activity_id

N - Adding data into tables

Many-to-Many Relationship between Activity and Material



Material_ID	Name_Spec	Category	Price	Curren	Min_	Purchase_Date	Expiry_Date	Status	Maintenance_Log	QRCode_Path	Supplier_ID
MAT-DEC0001	Table Linens-10W-Gold	Decoration	1668.61	86	14	2025/2/7	2027/8/1	Idle	2025-04-01 John Calibration	/qrcode/12e8f40e-ea3fSUP-005	
MAT-DEC0002	Chiavari Chair-5W-Black	Decoration	4819.12	77	12	2024/9/25	2028/1/19	Idle	2025-04-01 John Calibration	/qrcode/a74c5a1f-eb1fSUP-012	
MAT-EQU0003	Projector-10W-Gold	Equipment	459.69	63	8	2024/6/7	2027/12/13	Reserved	2025-04-01 Bob Cleaning	/qrcode/db34c43d-1bd4SUP-005	
MAT-EQU0004	Projector-5W-White	Equipment	4477.22	43	20	2025/3/2	2027/6/15	Out	2025-04-01 Bob Inspection	/qrcode/c641d589-4604SUP-001	
MAT-LIG0005	Laser System-20W-Black	Lighting	2187.43	73	10	2024/7/5	2026/6/26	Out	2025-04-01 John Inspection	/qrcode/5e08c8ba-078fSUP-012	
MAT-EQU0006	Sound System-10W-White	Equipment	310.9	92	6	2024/6/30	2027/9/18	Idle	2025-04-01 Bob Inspection	/qrcode/d72b29f8-4a0fSUP-010	
MAT-LIG0007	Laser System-20W-White	Lighting	4472.13	62	7	2024/7/25	2028/3/24	Idle	2025-04-01 Bob Cleaning	/qrcode/08c6e913-663fSUP-012	
MAT-EQU0008	Projector-10W-Gold	Equipment	4647.77	62	11	2024/11/20	2026/6/7	Idle	2025-04-01 Bob Cleaning	/qrcode/09597273-114fSUP-015	
MAT-EQU0009	LED Screen-5W-Gold	Equipment	4130.71	40	12	2025/1/6	2027/7/11	Reserved	2025-04-01 Alice Calibration	/qrcode/2b5b7292-dad4SUP-010	
MAT-EQU0010	Projector-20W-Black	Equipment	2743.27	94	20	2024/7/14	2028/2/8	Idle	2025-04-01 Bob Cleaning	/qrcode/dfe5bc8a-ee8fSUP-015	

Material

Activ Name	Type	Start_Date	End_Date	Expect	Actual_	Status	Descrip	Venue_ID	FinancialID
251 Tech Company Event	Conference	2024/1/7	2024/6/19	16	31	Cancelled	None	V003	FIN-2025-0001
252 E-commerce Group Event	Training	2024/2/29	2024/2/29	20	84	Cancelled	None	V004	FIN-2025-0002
253 Pharmaceutical Company Event	Conference	2024/4/23	2024/5/28	76	95	Cancelled	None	V005	FIN-2025-0003
254 Catering Group Event	Exhibition	2024/3/1	2024/8/1	26	36	Cancelled	None	V003	FIN-2025-0004
255 Construction Company Event	Workshop	2024/2/16	2024/5/24	65	27	Cancelled	None	V005	FIN-2025-0005
256 IT Company Event	Workshop	2024/5/18	2024/1/11	91	97	Ongoing	None	V002	FIN-2025-0006
257 Education Technology Company Event	Conference	2024/2/15	2024/4/14	24	88	Cancelled	None	V005	FIN-2025-0007
258 Retail Company Event	Training	2024/10/17	2024/10/18	73	80	Scheduled	None	V003	FIN-2025-0008
259 Advertising Company Event	Exhibition	2024/8/9	2024/4/1	79	53	Scheduled	None	V005	FIN-2025-0009

Activity

Activity	MaterialID
251	MAT-DEC0001
251	MAT-EQU0011
251	MAT-EQU0012
251	MAT-LIG0013
251	MAT-EQU0004
252	MAT-DEC0002
253	MAT-EQU0003
254	MAT-EQU0004
255	MAT-LIG0005

Activity_needs_material


```
SELECT A.Activity_ID, A.Name, A.Type, A.Start_Date, A.End_Date, V.Venue_Name, V.Venue_Type, F.Amount, F.Budget_Category
FROM Activity A
JOIN venue V ON A.Venue_ID = V.Venue_ID
JOIN Financial F ON A.FinancialID = F.FinancialID;
```

Result

	Activity_ID	Name	Type	Start_Date	End_Date	Venue_Name	Venue_Type	Amount	Budget_Category
▶	256	IT Company Event	Workshop	2024-05-18 00:00:00	2024-01-11 00:00:00	Chengdu Exhibition Hall	Exhibition Hall	4437.96	Decoration
	267	Performing Arts Company Event	Exhibition	2024-12-30 00:00:00	2024-04-11 00:00:00	Chengdu Exhibition Hall	Exhibition Hall	1113.09	Venue Rental
	269	Telecom Company Event	Workshop	2024-05-26 00:00:00	2024-12-06 00:00:00	Chengdu Exhibition Hall	Exhibition Hall	20892.70	Catering
	251	Tech Company Event	Conference	2024-01-07 00:00:00	2024-06-19 00:00:00	Chengdu Creative Space	Training Room	16522.84	Staffing
	254	Catering Group Event	Exhibition	2024-03-01 00:00:00	2024-08-01 00:00:00	Chengdu Creative Space	Training Room	24784.26	Venue Rental
	258	Retail Company Event	Training	2024-10-17 00:00:00	2024-10-18 00:00:00	Chengdu Creative Space	Training Room	4647.77	Venue Rental
	262	Creative Design Company Event	Workshop	2024-06-17 00:00:00	2024-06-10 00:00:00	Chengdu Creative Space	Training Room	5907.10	Marketing
	264	Financial Company Event	Training	2024-05-30 00:00:00	2024-08-18 00:00:00	Chengdu Creative Space	Training Room	5005.83	Staffing
	266	Network Company Event	Workshop	2024-04-21 00:00:00	2024-04-14 00:00:00	Chengdu Creative Space	Training Room	71554.08	AV Equipment
	268	Culture Company Event	Workshop	2024-02-27 00:00:00	2024-04-25 00:00:00	Chengdu Creative Space	Training Room	74352.78	Staffing
	252	E-commerce Group Event	Training	2024-02-29 00:00:00	2024-02-29 00:00:00	Chengdu Arts Plaza	Main Hall	50142.48	Staffing
	261	Real Estate Company Event	Training	2024-02-05 00:00:00	2024-08-22 00:00:00	Chengdu Arts Plaza	Main Hall	16459.62	Venue Rental
	263	Environmental Protection Comp...	Workshop	2024-06-03 00:00:00	2024-10-10 00:00:00	Chengdu Arts Plaza	Main Hall	23780.28	
	265	Tourism Company Event	Exhibition	2024-09-08 00:00:00	2024-01-24 00:00:00	Chengdu Arts Plaza	Main Hall	49378.86	Staffing
	253	Pharmaceutical Company Event	Conference	2024-04-23 00:00:00	2024-05-28 00:00:00	Chengdu Tech Hub	Conference ...	56115.67	Venue Rental
	255	Government Company Event	Workshop	2024-03-16 00:00:00	2024-05-21 00:00:00	Chengdu Tech Hub	Conference ...	56058.87	Staffing

This query provides a comprehensive overview of each activity, including the venue details and its associated financial costs (amount and budget category). It is essential for event planning, budgeting, and reporting ,allowing the business to track expenses and ensure activities align with financial allocations.



- ```
SELECT Type, AVG(Rating) AS Avg_Rating
FROM Supplier
GROUP BY Type
HAVING AVG(Rating) > 4.0;
```

## Result

|   | Type     | Avg_Rating        |
|---|----------|-------------------|
| ▶ | Material | 4.35              |
|   | Service  | 4.158333333333334 |

Identifying supplier types with consistently high ratings helps the business prioritize reliable supplier categories for future collaborations. This ensures quality and efficiency in sourcing materials and services, critical for maintaining event standards.



# N - Query 3

QUERIES

Utilization



```
• SELECT M.Material_ID, M.Name_Spec, M.Category, M.Current_Stock, M.Min_Stock, S.Company_Name, S.Rating,S.Backup_Supplier_ID,BS.Rating as BS_Rating
FROM Material M
JOIN Supplier S ON M.Supplier_ID = S.Supplier_ID
JOIN Supplier BS ON BS.Supplier_ID = S.Backup_Supplier_ID
WHERE M.Current_Stock < M.Min_Stock;
```

## Result

|   | Material_ID | Name_Spec          | Category  | Current_Stock | Min_Stock | Company_Name          | Rating | Backup_Supplier_ID | BS_Rating |
|---|-------------|--------------------|-----------|---------------|-----------|-----------------------|--------|--------------------|-----------|
| ▶ | MAT-EQU0011 | Projector-5W-White | Equipment | 0             | 18        | GlobalEvents_58042668 | 4.2    | SUP-001            | 3.7       |

**This query highlights materials below their minimum stock levels, including supplier details for immediate reordering. And we grab rating of the company with inventory risk to make an initial assessment if we need to contact with backup supplier or other supplementary solutions if the credit of backup supplier still being not totally untrustworthy.**

# N - Query 4

QUERIES

Utilization



```
• SELECT A.Activity_ID, A.Name, E.Name AS Employee_Name, ED.Position
FROM Activity A
JOIN Activity_has_Employee AE ON A.Activity_ID = AE.Activity_ID
JOIN Employee E ON AE.Employee_ID = E.Employee_ID
JOIN Regular_Employee ED ON E.Employee_ID = ED.Employee_ID;
```

## Result

|   | Activity_ID | Name                         | Employee_Name | Position             |
|---|-------------|------------------------------|---------------|----------------------|
| ▶ | 251         | Tech Company Event           | David Wang    | General Manager      |
|   | 252         | E-commerce Group Event       | Jessica Li    | Marketing Director   |
|   | 253         | Pharmaceutical Company Event | Thomas Zhang  | CFO                  |
|   | 253         | Pharmaceutical Company Event | Amanda Chen   | Event Planner        |
|   | 253         | Pharmaceutical Company Event | Robert Liu    | Head Chef            |
|   | 253         | Pharmaceutical Company Event | Sarah Zhang   | HR Manager           |
|   | 253         | Pharmaceutical Company Event | Daniel Chen   | Operations Manager   |
|   | 251         | Tech Company Event           | Emily Wang    | Junior Designer      |
|   | 253         | Pharmaceutical Company Event | Emily Wang    | Junior Designer      |
|   | 251         | Tech Company Event           | Michael Zhang | Senior Accountant    |
|   | 253         | Pharmaceutical Company Event | Michael Zhang | Senior Accountant    |
|   | 251         | Tech Company Event           | Rachel Li     | Marketing Specialist |
|   | 253         | Pharmaceutical Company Event | Rachel Li     | Marketing Specialist |
|   | 251         | Tech Company Event           | Samuel Wang   | Executive Chef       |
|   | 253         | Pharmaceutical Company Event | Samuel Wang   | Executive Chef       |

**Understanding the team composition for each activity is crucial for resource allocation and workload management. This query shows which employees are assigned to each activity and their roles, helping managers ensure the right skills are in place and avoid overburdening staff.**



```
• SELECT Activity_ID, Name, Actual_Participants
FROM Activity
WHERE Actual_Participants > (SELECT AVG(Actual_Participants) FROM Activity);
```

Result

|   | Activity_ID | Name                               | Actual_Participants |
|---|-------------|------------------------------------|---------------------|
| ▶ | 252         | E-commerce Group Event             | 84                  |
|   | 253         | Pharmaceutical Company Event       | 95                  |
|   | 256         | IT Company Event                   | 97                  |
|   | 257         | Education Technology Company Event | 88                  |
|   | 258         | Retail Company Event               | 80                  |
|   | 260         | Logistics Company Event            | 79                  |
|   | 264         | Financial Company Event            | 100                 |
|   | 268         | Culture Company Event              | 94                  |
|   | 269         | Telecom Company Event              | 89                  |
|   | 270         | Automobile Company Event           | 68                  |
| ⌵ | NULL        | NULL                               | NULL                |

Identifying activities with higher-than-average attendance helps the business analyze what makes events successful. This insight can guide future planning and marketing strategies to replicate high-turnout events, enhancing customer engagement and satisfaction.

## - An automatic activity planner based on LLM

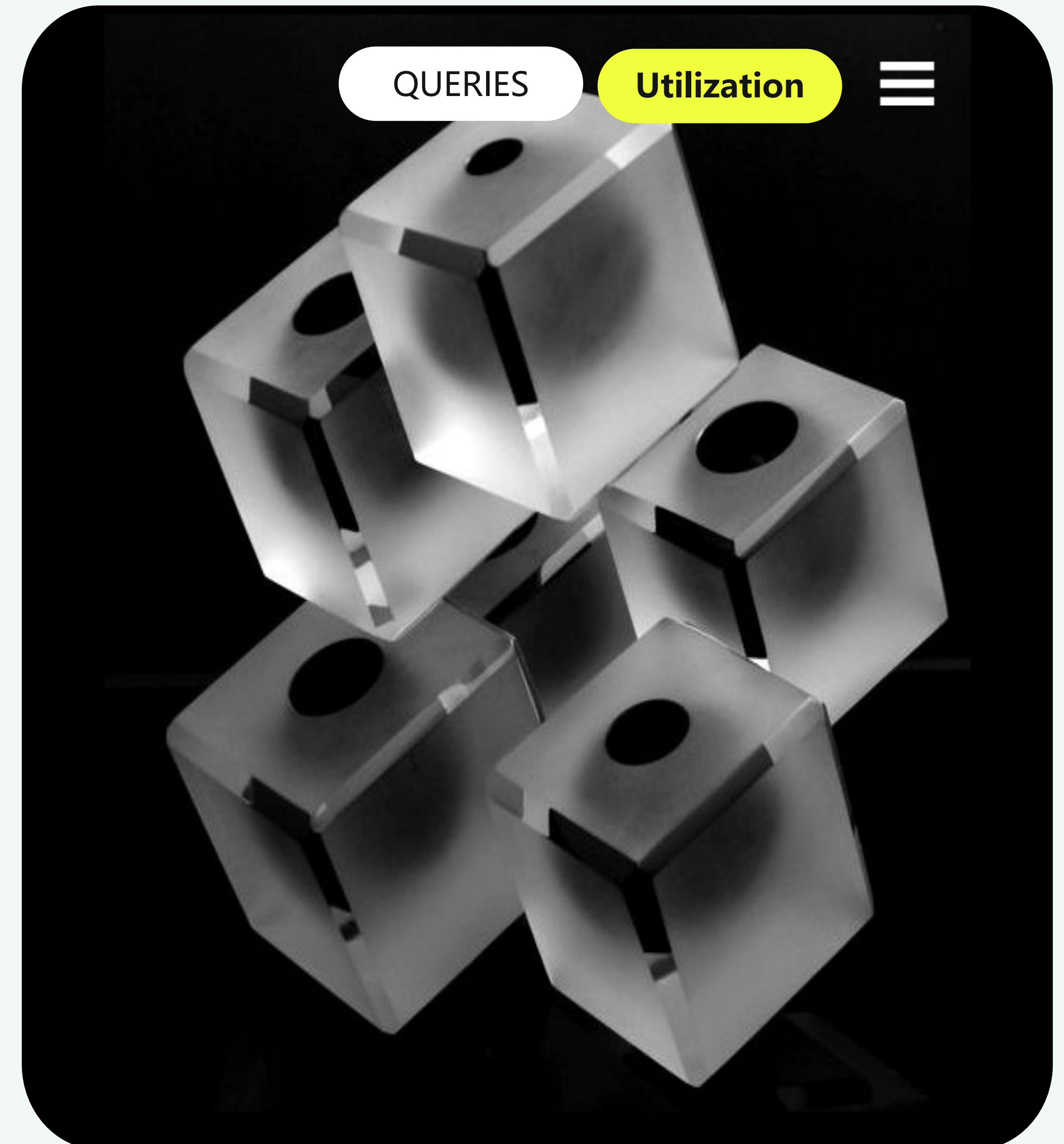
To ensure the constructed database maintains practical relevance, we developed an automated event scheduling model based on **localized LLM** technology. This model can provide event arrangement recommendations by processing direct **natural language instructions**.

While this implementation may appear redundant given the current small-scale dataset, the system demonstrates its full potential **in large-volume, multi-constraint scenarios** where it can perform comprehensive event planning.

INTRODUCTION

BUSSINESS MODEL

ACHIEVEMENT







## Venue

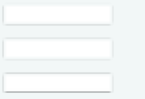
## Materials & Supplier

## Employee

```
cursor.execute("""
 SELECT V.Venue_Name, V.Venue_Type, V.Location, V.Capacity, V.Rent_Per_Hour, V.Contact_Info
 FROM venue V
 WHERE V.Availability = 'Available'
 AND V.Venue_avalable_date <= %s
 AND V.Venue_Type LIKE %s
 AND V.Capacity >= %s;
""", (date_to_arrange, f"%{activity_type}%", capacity))
```

```
Query materials
cursor.execute("""
 SELECT m.Material_ID, m.Name_Spec, m.Category, m.Price, m.Current_Stock, s.Supplier_ID, s.Company_Name, s.Rating, s.
 Backup_Supplier_ID
 FROM Material AS m
 JOIN Supplier s ON s.Supplier_ID = m.Supplier_ID
 WHERE m.status = 'Idle'
 AND m.Current_Stock > 0
 AND (m.Name_Spec LIKE %s OR m.Name_Spec LIKE %s OR m.Name_Spec LIKE %s);
""", (f"%{material_1}%", f"%{material_2}%", f"%{material_3}%"))
```

```
Query employees
cursor.execute("""
 SELECT E.Employee_ID, E.Name, ED.Position, ED.Years_Of_Service
 FROM Employee E
 JOIN Regular_Employee ED ON E.Employee_ID = ED.Employee_ID
 WHERE ED.Years_Of_Service >= 3;
""")
```



```
Create prompt for the LLM
p1 = f"A client plans to hold a {activity_type} on {date_to_arrange}, which needs a capacity of {capacity} and lasts for {hour} hours."
p2 = "Available venue information:\n" + venues_str
p4 = "Supplier information:\n" + suppliers_str
p5 = "Available employee information:\n" + employees_str
p6 = f"### Budget combination options ({len(budget_combinations_str.split('###'))} types):\n{budget_combinations_str}"
p7 = "Please provide the best activity arrangement plan based on the above information. Please be specific. Output the result in Markdown format."
p8 = 'Summarize all the arrangements in one sentence separately at the end of the answer starting with $$$$'
prompt = '\n'.join([p1, p2, p4, p5, p6, p7, p8])

Call DeepSeek model
stream = chat(model="deepseek-r1:1.5b", messages=[{"role": "user", "content": prompt}], stream=False)
prediction_content = stream["message"]["content"].strip()
```





# Activity Planner

Date (YYYY-MM-DD):

mm/dd/yyyy

Activity Type:

Capacity:

Hours:

Material 1:

Quantity:

Material 2:

Quantity:

Material 3:

Quantity:

Generate Plan

|                                                                                                              |  |
|--------------------------------------------------------------------------------------------------------------|--|
| ### Venue: Chengdu International Convention Center                                                           |  |
| - Cost of venue: 800.00 (4 hours × 200.00/hour)                                                              |  |
| - Combination of materials: LED Screen-10W-Gold × 1, Sound System-10W-White × 1, Chiavari Chair-5W-Black × 5 |  |
| - Cost of materials: 28221.70                                                                                |  |
| - Total cost: 29021.70                                                                                       |  |
| ### Venue: Chengdu International Convention Center                                                           |  |
| - Cost of venue: 800.00 (4 hours × 200.00/hour)                                                              |  |
| - Combination of materials: LED Screen-10W-Gold × 1, Sound System-10W-White × 1, Chiavari Chair-5W-Gold × 5  |  |
| - Cost of materials: 16813.50                                                                                |  |
| - Total cost: 17613.50                                                                                       |  |
| ### Venue: Chengdu International Convention Center                                                           |  |
| - Cost of venue: 800.00 (4 hours × 200.00/hour)                                                              |  |
| - Combination of materials: LED Screen-10W-Gold × 1, Sound System-20W-Gold × 1, Chiavari Chair-5W-Black × 5  |  |
| - Cost of materials: 32227.39                                                                                |  |
| - Total cost: 33027.39                                                                                       |  |
| ### Venue: Chengdu International Convention Center                                                           |  |
| - Cost of venue: 800.00 (4 hours × 200.00/hour)                                                              |  |
| - Combination of materials: LED Screen-10W-Gold × 1, Sound System-20W-Gold × 1, Chiavari Chair-5W-Gold × 5   |  |
| - Cost of materials: 20819.19                                                                                |  |
| - Total cost: 21619.19                                                                                       |  |

LLM is selecting the best arrangement for you, please wait...

## Final Arrangement

Okay, so I'm trying to figure out the best way for this client to hold a 4-hour conference on April 15th at Chengdu International Convention Center. The venue has a capacity of 500 and costs \$800 per hour. The budget is \$800, which would cover four hours at \$200 each.

First, I need to consider the client's needs beyond just the capacity. They want a conference that can accommodate up to 200 people. That seems manageable within the venue since it's over 500 seats. However, maybe there are some additional requirements like specific speaker sessions or materials needed. The client also has employee information from 19 employees, with different years of service ranging from 3 to 9 years.

Looking at the vendors, TechStage\_32285404 has ratings of 3.7, 4.7, and 4.4. GlobalEvents seems more reliable as they have two high ratings (4.4s). Each vendor's details are spread out with duplicates, but I can check each one.

For venue cost, the basic option is \$800, which covers four hours. So that part is covered. Now, for materials, there are several options provided. The client needs LED screens, sound systems, and chairs. Let me see if these are included in any of the vendor packages.

Looking at the first venue option, they include a screen, sound system, and gold chairs. Each chair set is 5W-Black, which might not be necessary if chairs aren't required as gold. If just chairs aren't needed, that cost might be too high. So maybe the second option is better because it's more budget-friendly: one screen, one sound system in gold, and chairs that are probably fine.

I should check each vendor for the specific components needed. The materials list includes LED screens (10W-Gold), Sound System (both 10W-Gold and 20W-Gold). For sound, if they don't need professional speakers, the 10W might be better. The chair sets are in different colors: one gold and four black. Since we're focusing on chairs, maybe the black ones would suffice.

Looking at the employee info, the client has employees from ages 3 to 9 years. If the event requires staff to be present for all activities, perhaps having a full team or some key personnel assigned is necessary. I should note if any of the vendors have teams that can handle the required number of employees. For example, if there are specific roles like speakers or coordinating team, those would need to be included.

Also, considering the budget: each hour at \$200 is manageable with a total of \$800 for four hours. The client doesn't mention any additional staff or materials beyond capacity and basic equipment, so I think sticking with the venue and the most cost-effective materials option is best.

I should summarize all these points to present a clear plan that meets the client's needs within their budget.

The client plans to hold a 4-hour conference at Chengdu International Convention Center with a capacity of 500, which provides ample space for up to 200 participants. The venue cost of \$800 covers four hours at \$200 per hour. To meet the requirements, including speaker sessions and materials, we recommend using the "Basic" venue option with the following equipment:

- **Materials:** LED Screen-10W-Gold × 1, Sound System-10W-Gold × 1, and Chiavari Chair-5W-Black × 5. This arrangement includes all necessary components for the event while staying within budget.

## Summary



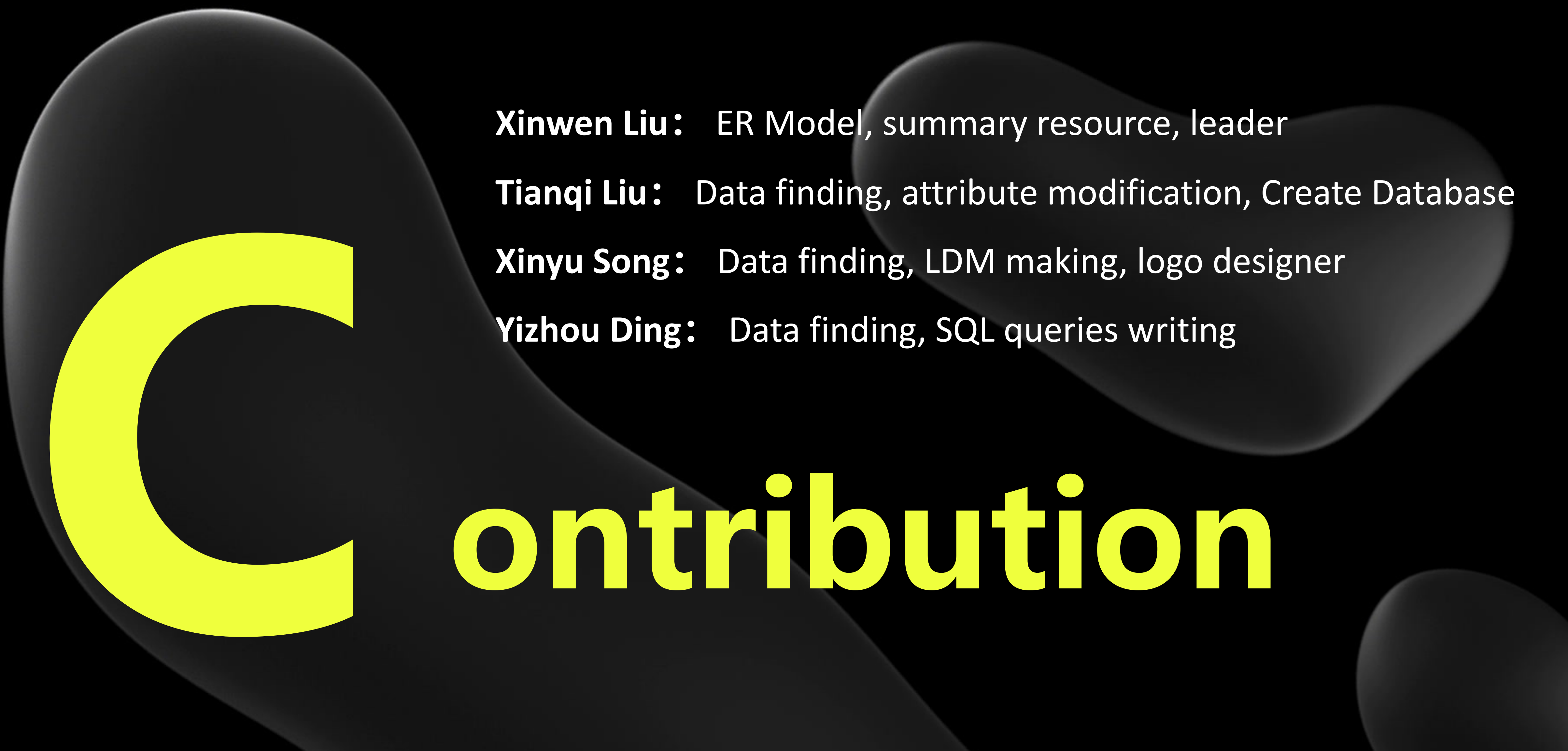


**Xinwen Liu:** ER Model, summary resource, leader

**Tianqi Liu:** Data finding, attribute modification, Create Database

**Xinyu Song:** Data finding, LDM making, logo designer

**Yizhou Ding:** Data finding, SQL queries writing

A large, bright yellow letter 'C' is positioned on the left side of the page. The background is dark with several large, semi-transparent, rounded shapes in shades of gray and black, creating a modern, abstract design.

# ontribution





*SELECT \* FROM happiness  
WHERE venue=*

**'Nectar'**